

Sagar Barvaliya

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Highly accomplished Data Engineer with extensive experience in building scalable data pipelines and data warehousing solutions, demonstrated by leading the Mach1ML project migration to GCP and implementing automated CI/CD pipelines. Proven ability to enable analytics and business intelligence through data modeling and AI workflow implementation.

SKILLS

Languages:	JavaScript, Typescript, Java, Python, C++, Go
Frontend Technologies:	TypeScript, React, Angular, Next.js, JavaScript, Redux, Mobx, HTML, CSS, Tailwind, jQuery
Backend:	Python, FastAPI, Node.js, Spring Boot, Java, J2EE, Servlet, JMS, Hibernate, JPA, Spring MVC
Data:	PostgreSQL, Oracle SQL, MySQL, MongoDB, DynamoDB, S3, Redis, Kafka
Frameworks/Concepts:	Express.js, JWT, Validator.js, WebSockets, Promises, Generators, REST APIs, OAuth2, OIDC
DevOps:	AWS, Docker, Kubernetes, CI/CD, Tekton, Jenkins, GitHub, GCP, Terraform, Airflow
AI/ML:	TensorFlow, PyTorch, scikit-learn, Pandas, NumPy, MLflow, GitHub Copilot
Others:	JIRA, Documentation, Rally, Webex, Slack, Data Warehouse, Structured Data, Unstructured Data, Business Intelligence, AI Workflows, ETL

WORK EXPERIENCE

SENIOR SOFTWARE ENGINEER, FORD, Michigan, USA

Mar 2022 – Aug 2024

- Led the implementation of **Tekton Pipelines** and **shell scripts** for automated CI/CD, resulting in a 60% reduction in manual deployment effort and enabling faster, more **frequent releases**. This significantly improved team efficiency and accelerated software delivery, aligning with a fast-paced **startup environment**.
- Engineered robust data models and ETL processes using **SQL** and **Python**, resulting in a 35% improvement in data processing speed and enabling more efficient analytics for business intelligence dashboards.
- Built and **automated Docker** images, saving 50+ hours monthly and streamlining the release process. Developed **Airflow** projects, cutting pipeline setup time by 40% and enhancing project scalability and execution speed.
- Led the **migration** of the **Mach1ML** project to **Google Cloud Platform**, resulting in a 30% reduction in project maintenance costs. This provided a more scalable and reliable hosting environment and demonstrates experience with **cloud infrastructure**.
- Implemented **FOSSA** and **SonarQube**, reducing security risks by 20% and improving code cleanliness, which led to fewer defects and more secure software development. Actively using **GitHub Copilot** to improve code quality and accelerate development.
- Designed and implemented an **MLOps** pipeline using **MLflow** and **Kubernetes** to automate the training, deployment, and monitoring of **machine-learning models**. This resulted in a 40% reduction in the time to deploy new models.
- Maintained comprehensive **documentation**, reducing onboarding time for new developers by 25% and enhancing team **knowledge sharing**. Collaborated closely with product owners to define user stories and deliver features aligned with product vision.

JAVA FULL STACK DEVELOPER, DYNAMIC WEB SOFT, SURAT, INDIA

Oct 2018 - Dec 2019

- Designed and implemented **REST-based microservices** using **Spring Boot**, resulting in a 30% improvement in system flexibility and the ability to handle **1 million+ API** requests daily.
- Designed and developed **EcoTrack**, an **IoT-based** environmental monitoring system using **Node.js** and **Angular**. Built **Angular 6.0** single-page applications, reducing page load times by 50% and boosting user engagement by 25%. Utilized **Java 8** features for 30% faster data processing and simplified code complexity.
- Designed and implemented a comprehensive data warehousing solution using **PostgreSQL** and **AWS S3**, improving data accessibility by 60% and enabling stakeholders to derive actionable insights.
- Deployed and managed **AWS** services such as **EC2, VPC, S3, Lambda**, achieving 99.9% uptime and reducing downtime, while using **MongoDB** and **Spring MVC** to improve data retrieval efficiency by 40%.
- Refactored code for **JVM** optimization, reducing infrastructure costs by 15% and improving application speed by 25%, while monitoring **AWS** with **CloudWatch**, cutting downtime by 15%.
- Utilized **C++** to optimize core backend services, significantly improving data processing and retrieval times by 30%, contributing to the scalability and performance of **microservices architecture**. Developed **gRPC** interfaces for improved communication and reducing response latency.
- Collaborated with Business Analysts to gather requirements, create **data flow diagrams**, and design **UML** models, improving project clarity and reducing scope changes.
- Developed and scaled data pipelines leveraging **Kafka** and **Airflow** to ingest and process large volumes of unstructured data, resulting in a 40% reduction in data latency and improved real-time analytics capabilities.
- Developed **HealthHub**, a full-stack web application for **healthcare management** using **AngularJS** and **Java**. Built responsive web interfaces using **Struts MVC, JSP, AngularJS, and Bootstrap**, enhancing user experience and improving performance.
- Developed a robust persistence layer with **DAO** and **Hibernate**, and improved data handling and retrieval efficiency by over 30%.
- Implemented AI workflows using **TensorFlow** and **PyTorch** within a **Kubernetes** environment, improving model training efficiency by 50% and accelerating the deployment of AI-driven features for healthcare management.
- Leveraged **C** and **C++** for maintaining legacy systems, developing **real-time applications**, and integrating **low-level hardware communication**, resulting in a 40% improvement in system response time and enhanced overall system stability.

EDUCATION

ILLINOIS STATE UNIVERSITY, Normal, IL - Masters in Computer and Information Science

Government Engineering College, Gandhinagar, Gujarat - Bachelors in Computer Engineering